

Physiology of Thermoregulation

Olga Savinova

Biomedical Sciences

osavinov@nyit.edu

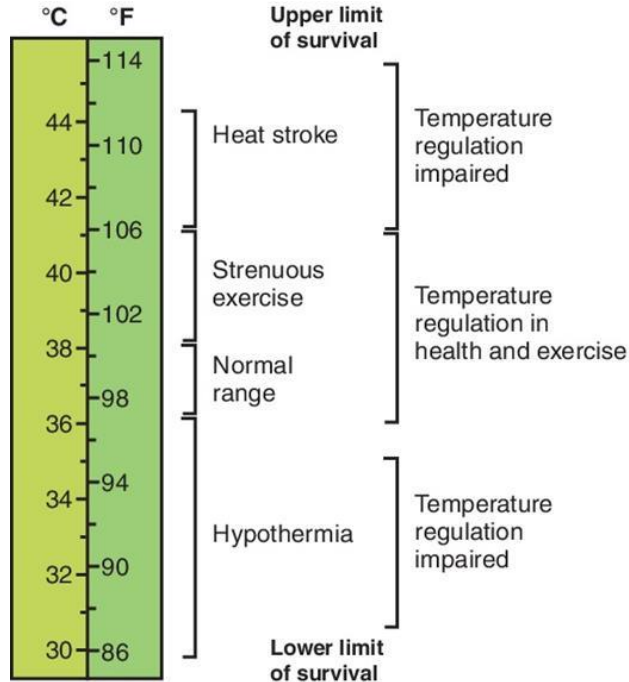
Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

At the conclusion of the session, the student should be able to:

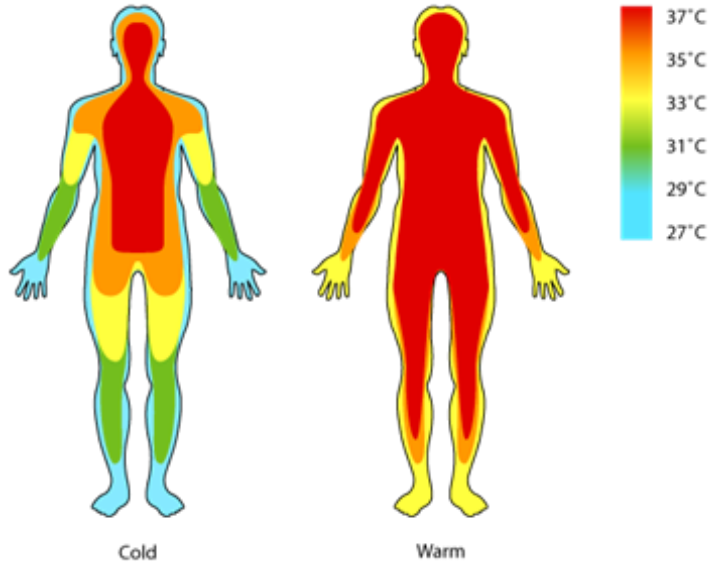
1. Compare and contrast core body temperature and skin temperature under different conditions
2. Describe the homeostatic mechanisms of maintaining stable core body temperature, including heat production and dissipation
3. Understand how the central nervous system regulates body temperature
4. Describe the mechanism of fever

Survival ranges of body temperatures



- Internal body temperatures are regulated within a narrow range, despite wide variations in environmental temperature
- Failure of thermoregulatory mechanisms leads hyper- or hypothermia
- Extreme levels of hyper and hypothermia limit survival

Core and Skin Temperature

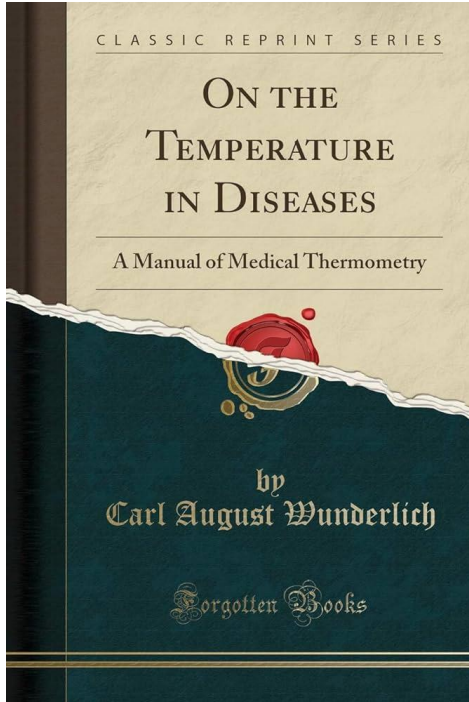


- **Core Temperature** – remains stable
 - Protects the function of internal organs
- **Skin Temperature** – rises and falls with surrounding temperature

Normal body temperature

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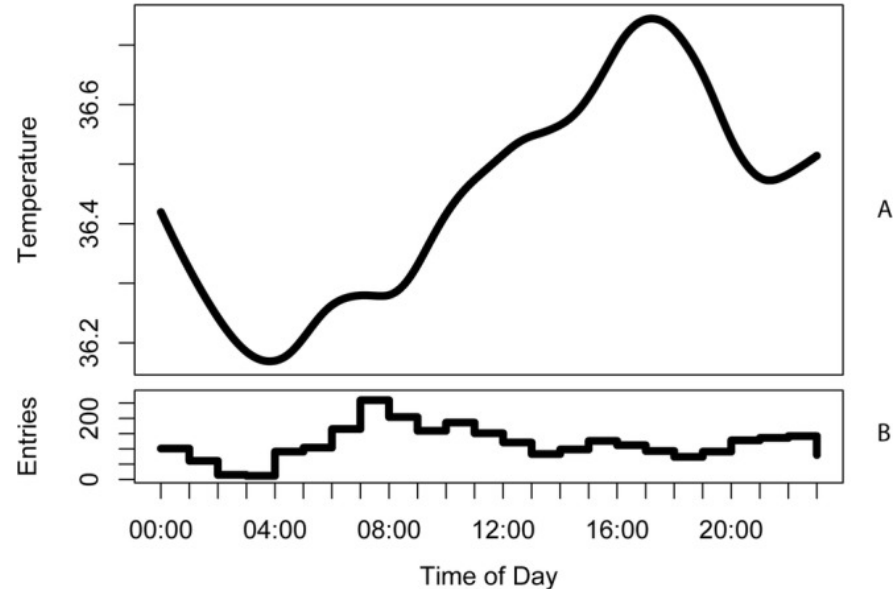
- For a long time, the temperature of 98.6°F (37C) – was considered “normal”
- In 1868, German physician Carl Wunderlich reported this figure for the first time as an average temperature in his rather large study population
- Average body temperature in the U.S. has dropped from 98.6F by about 0.05 F every decade since then
- Today’s average body temperature is closer to 97.9 F, with a range of 97.3°F to 98.2°F, when measured orally in healthy adult subjects

Body temperature varies during the day

- Body temperature is lowest in the early morning and increases by 0.9-1.8°F by late afternoon
 - Minimum temperature - between 3 am and 5 am
 - Maximum temperature - between 4 pm and 6 pm
- A **morning** oral temperature of greater than **98.9°F** could be considered a fever, whereas a fever in the afternoon (and overall) is defined as **99.9°F**

JAMA 1992;268(12):1578-80.

Daily Temperature Trend



J Gen Intern Med. 2018 Dec;33(12):2046-2047.

Using Smartphone Crowdsourcing to Redefine Normal and Febrile Temperatures in Adults: Results from the Feverprints Study

Other factors affecting body temperature

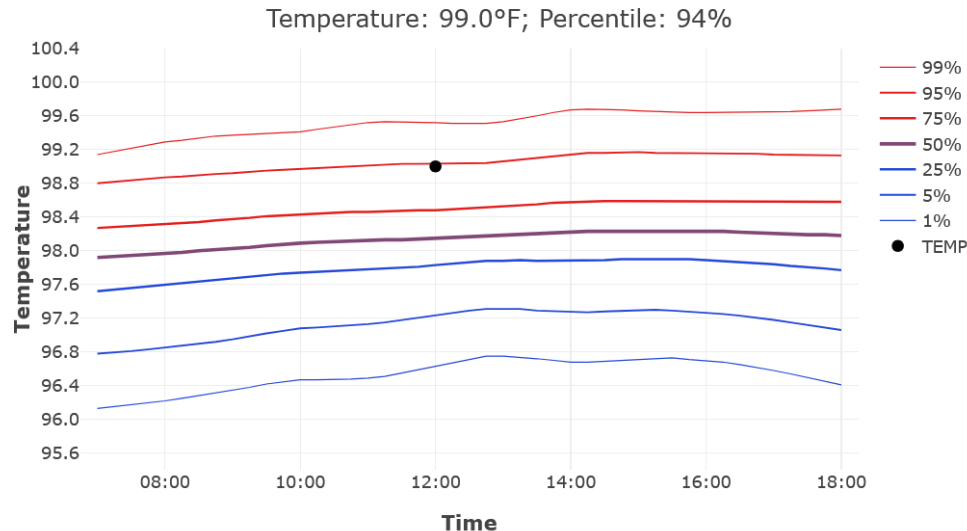
Personalized Temperature Ranges

This tool allows you to determine whether your oral temperature is within the usual range for someone like you, based on your entered height, weight, sex, and time of day (between 7am and 6pm).

A black dot indicates where your temperature lies within the usual range for someone like you. The central line (50%) is your average by time of day. As the lines get further from the middle your temperature is moving further from your average. A dot placed outside the top line (99th percentile) or bottom line (1st percentile) would be an unusual temperature for someone like you.

- Sex
 - higher in women
- Age
 - decreases with age
- Weight
 - increases with weight
- Height
 - decreases with height

Sex	Age	
☒ Female ☐ Male	30	
Size Unit	Height (ft)	Weight (lb)
☐ Metric ☒ US	5	168
	(in)	
	4	
Temperature Unit	Temp (°F)	Time
☐ °C ☒ °F	99.0	12:00
Reset to average woman Reset to average man		



<https://normaltemperature.stanford.edu/>

Pediatric fever: when to call a doctor?

Pediatric

Age	Tympanic Temp, F
0 - 3 mo	98.5 – 99.4
3 mo - 2 yrs	98.5 – 97.7
2 - 10 yrs	98.3 – 98.5
> 10 yrs	98.1 – 98.4

2016 Pediatric Emergency Care 34(3)

Compare to Adults

Tympanic	97.9 - 98.0
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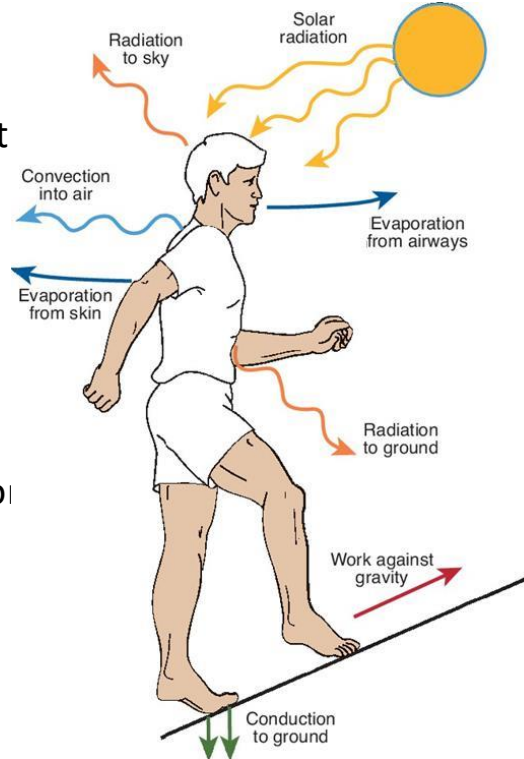
- Pediatric fever is defined as **100.4°F**

when to call a doctor?

- Newborn to 3 months old
>**100.4°F**
- 3 to 6 months old
> **102°F**
100.4 to 102°F and unusually irritable, lethargic, or uncomfortable
- 6 to 24 months old
> 102°F for longer than one day but no other signs
> 102°F and other signs (cold, cough, or diarrhea)
- Older than 2 years
Any fever lasting over 72 hours

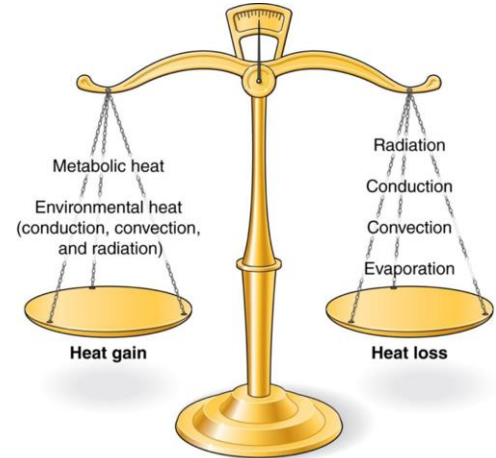
Heat exchange with the environment

- **Conduction**
 - Heat transfer through direct contact
- **Convection**
 - Heat transfer through the movement of air or water
- **Radiation**
 - Heat transfer through electromagnetic waves, without direct contact
- **Evaporation**
 - Heat transfer through the evaporation of water



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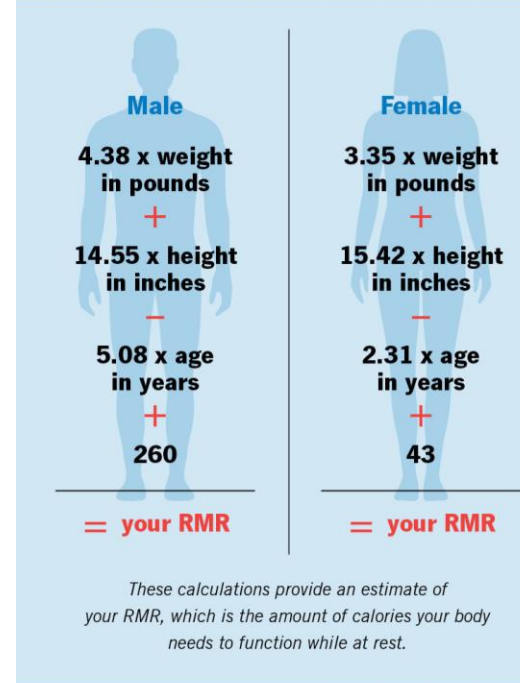
Basal/resting metabolic rate (BMR/RMR)

- Heat is produced as a byproduct of metabolism necessary to maintain chemical reaction in the brain and the internal organs
- BMR/RMR – metabolic rate under standard conditions at rest
 - 12 hour fast
 - Morning
 - 30 minutes rest
 - Comfortable temperature (~77F)
- BMR is measured by direct and indirect calorimetry
- BMR can be estimated based on the sex, weight, height, and age

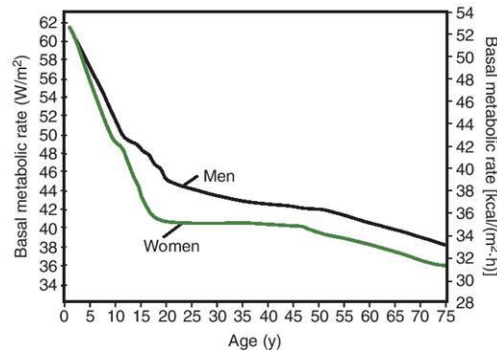
Region	% of body mass	% of heat produced at rest at ambient temp.
Brain	2	16
Trunk viscera	34	56
Muscle and skin	56	18
Other	8	10

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Effects of age and sex on the basal metabolic rate of healthy subjects



- Metabolic rate is proportional to body surface area
- Highest in infancy
- Increased in pregnancy
- Higher in males than females

Other factors affecting the metabolic rate

- Hormones
 - Catecholamines
 - Pheochromocytoma – weight loss, heat intolerance
 - Thyroxine
 - Hyperthyroidism - weight loss, heat intolerance
 - Hypothyroidism - weight gain, cold intolerance
 - Progesterone
 - Increase in body temperature after ovulation
- Thermic effect of food
 - 10-20% increase after meal
 - Fats and proteins more effective than carbohydrates

Heat production during external work/exercise

- Skeletal muscle becomes the main source of heat
- Metabolic rate can increase up to 10-fold
- Working muscles can warm circulating blood by 1-2°F

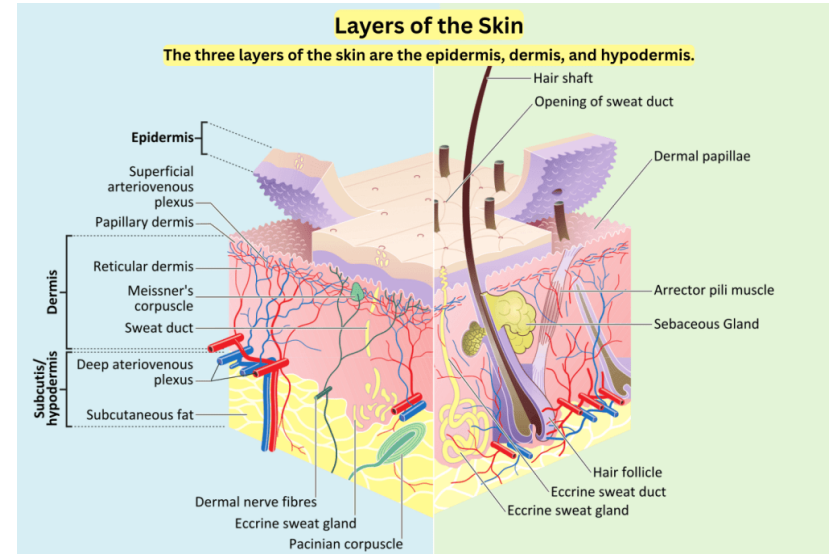
Region	% of body mass	% of heat produced at during exercise
Brain	2	1
Trunk viscera	34	8
Muscle and skin	56	90
Other	8	1

Thermoregulatory function of the skin

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- Skin and subcutaneous tissue provides insulation
- Blood flow to the skin from the body core provides heat transfer – dissipation of heat by radiation
- Activation of eccrine sweat glands increases heat escape by evaporation



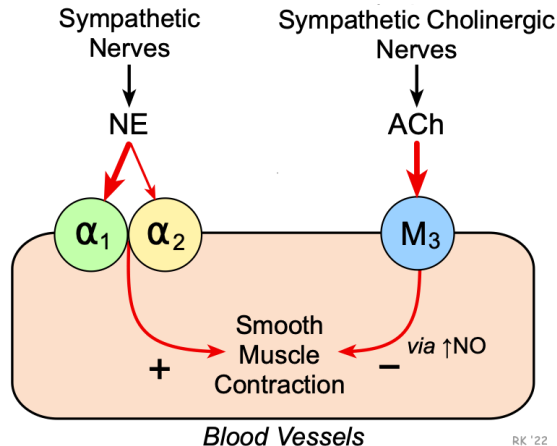
<https://sciencenotes.org/layers-of-the-skin-diagram-structure-function/>

Regulation of cutaneous blood flow

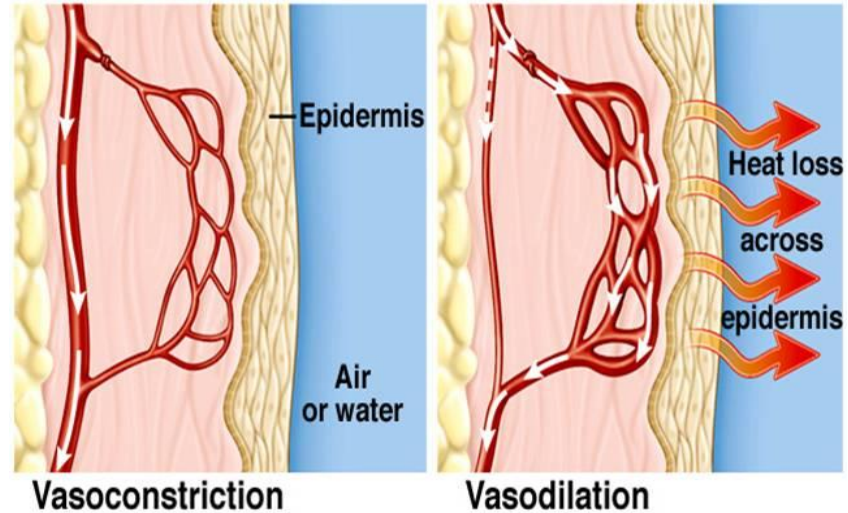
- Superficial arterio-venous plexus
- Sympathetic nervous system
 - adrenergic vasoconstriction
 - cholinergic vasodilation

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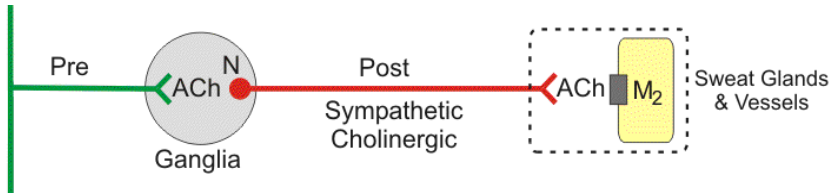


RK '22



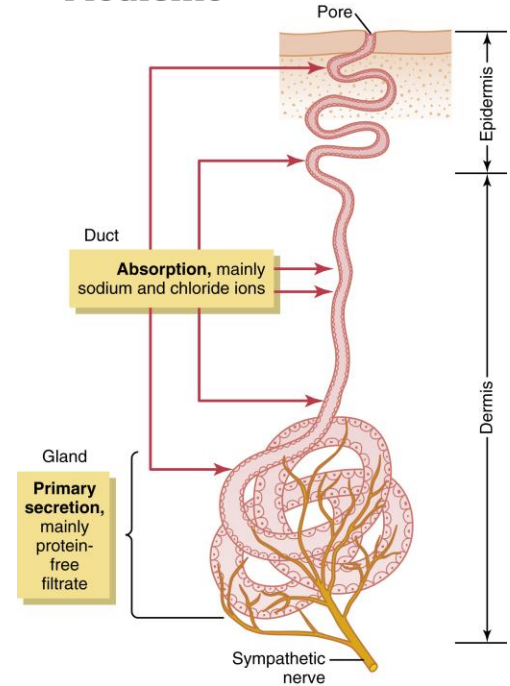
Mechanism of sweat production

- Eccrine glands of non-glabrous skin - face, trunk, and limbs
- Tubular structure consisting of two parts
 - deep subdermal coil
 - Produces primary secretion – composition of blood plasma minus the proteins
 - duct (in dermis and epidermis)
 - Reabsorbs sodium and chloride ions of the primary secretion
- The final composition of sweat depends on the rate of sweat production and flow, which is in turn is determine by the level of (cholinergic) sympathetic activation
- NOTE: Sweat glands are innervated by the post-ganglionic **cholinergic sympathetic** neurons



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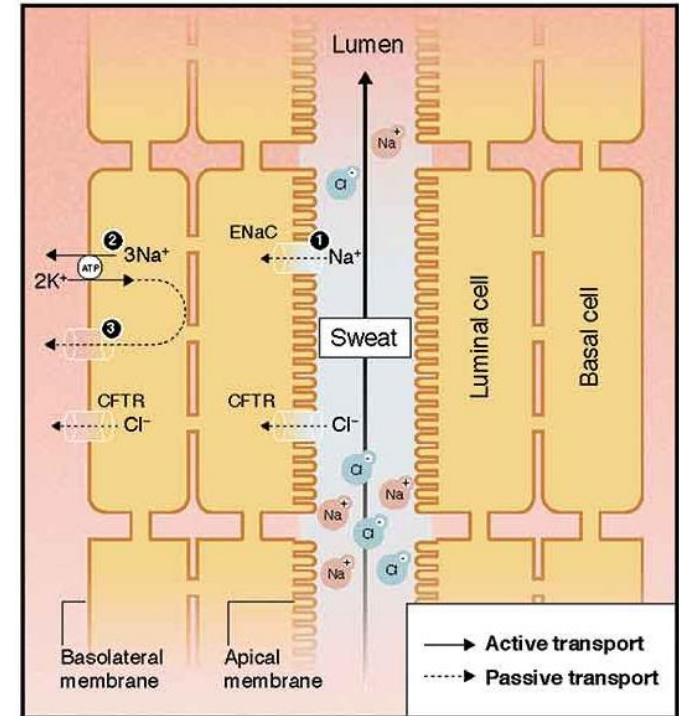
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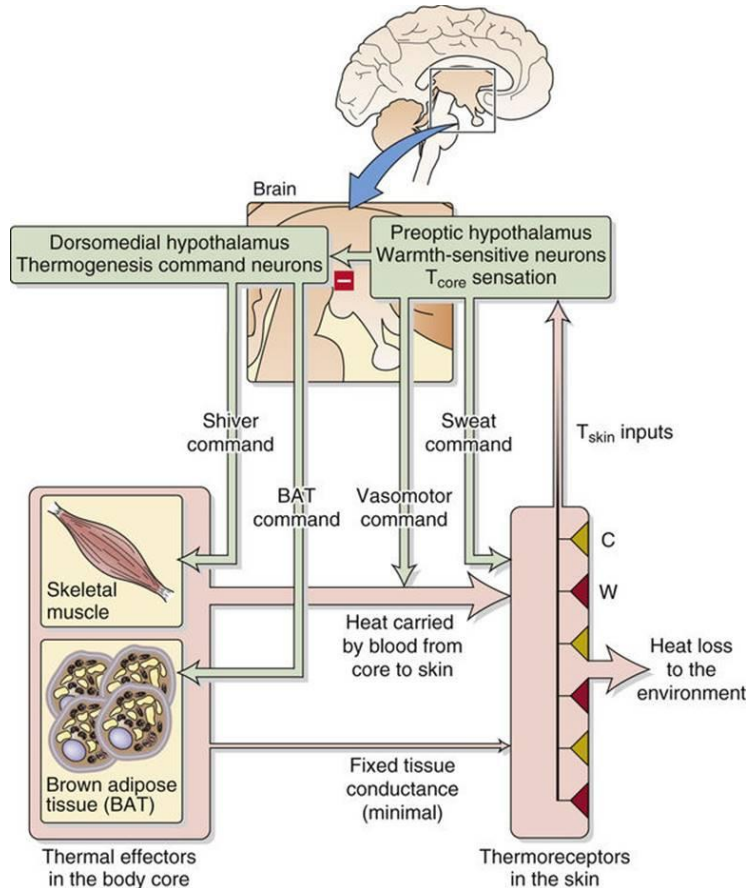
Acclimation to heat—the role of aldosterone

- Excessive sweating leads to the reduction of plasma volume and activation of **aldosterone** production by the adrenal gland
- Aldosterone acts on the kidney
 - increases sodium and water reabsorption and increases plasma volume
- Aldosterone acts on the luminal cells of the sweat gland
 - upregulates epithelial sodium channel (ENaC), the first step in sodium reabsorption from the sweat

An unacclimated person can lose 15 to 30 grams of salt each day. After 4 to 6 weeks, the loss is 3 to 5 g/day



Thermal Sensing



- **Skin Thermoreceptors**
 - Exposed to body surface temperature
 - More sensitive to cold
 - Greater number of cold than warmth receptors
- **Hypothalamic Temperature-Sensitive Neurons**
 - The principal mechanism for sensing core temperature
 - Set the temperature threshold
 - Detect deviation of temperature from the threshold

Effector Mechanisms

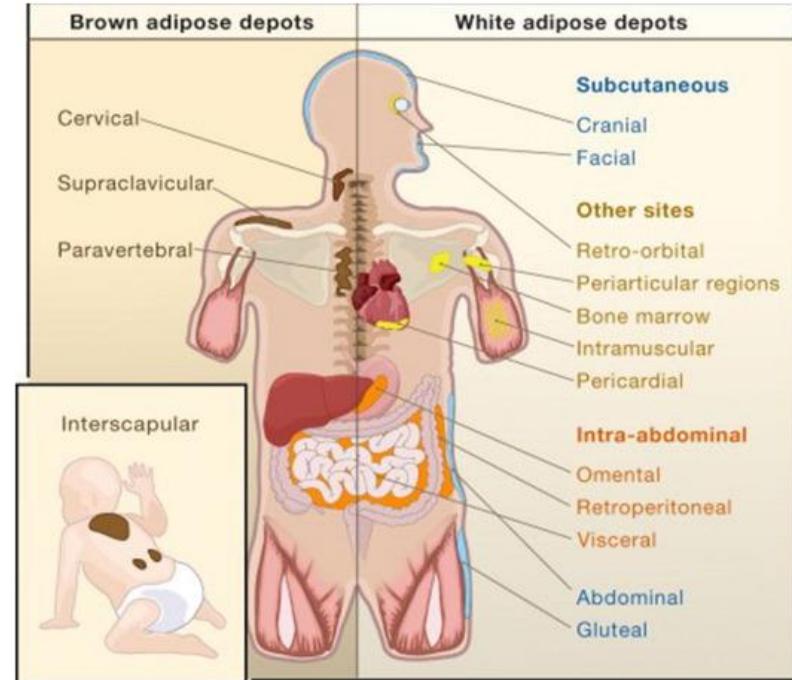
- Behavior
 - changing the environment
 - adding or removing clothing or other protective layering
- Temperature decreasing
 - Vasodilation of skin blood vessels
 - Sweating
- Temperature increasing
 - Vasoconstriction of skin blood vessels
 - Shivering
 - Chemical Thermogenesis

Chemical Thermogenesis

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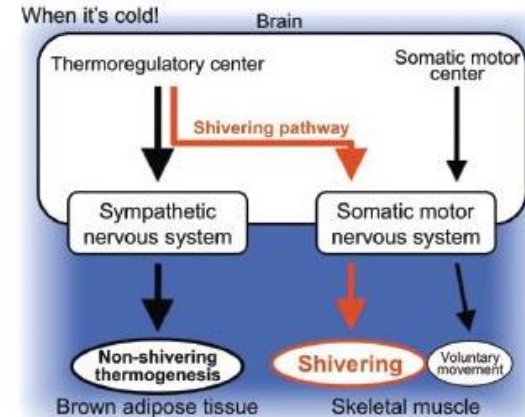
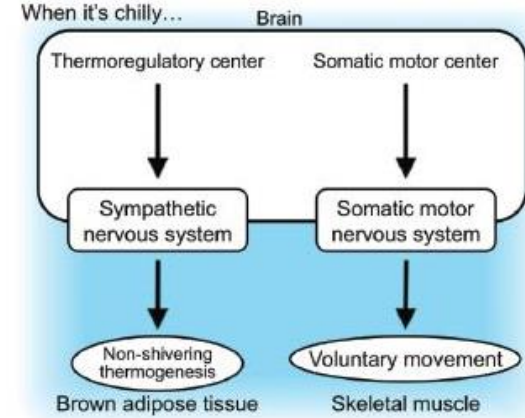
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- Brown adipose tissue
- Abundant lipid droplets and mitochondria
- Sympathetic innervation
- Activates Lipolysis
- Mitochondria uncoupling protein 1 (UCP1)
- Uncoupling of electrochemical proton gradient and ATP synthesis
- Energy dissipates as heat



Shivering thermogenesis

- increase in basal skeletal muscle tone
- involuntary, clonic, rhythmic contractions
- motor neurons innervation
- Impaired voluntary movements



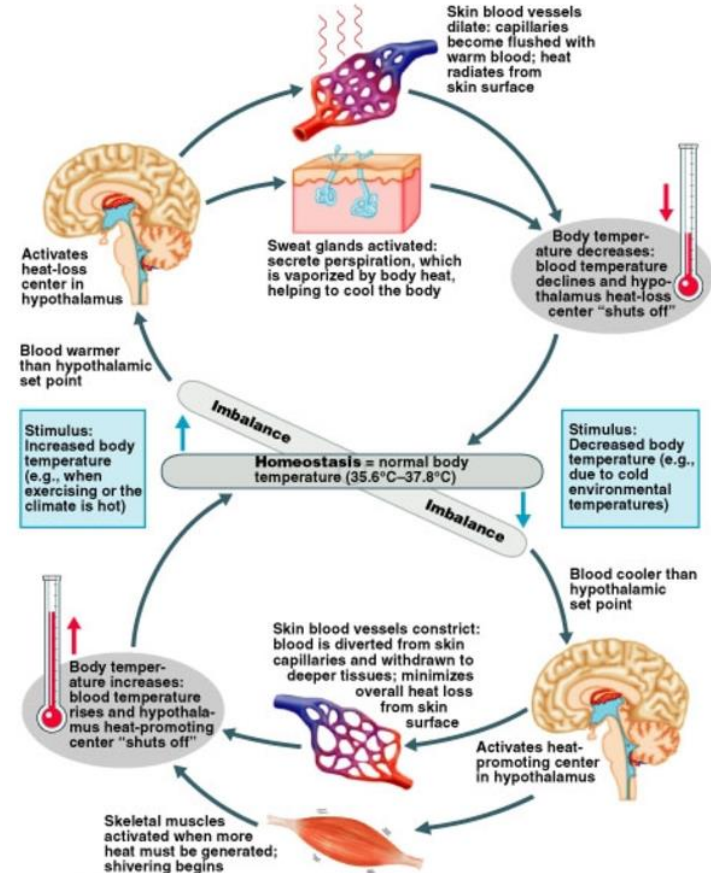
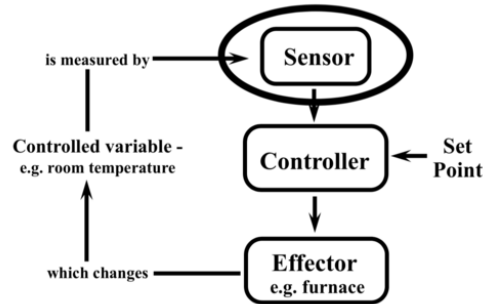
Hierarchical thermoregulatory response

- Behavioral Responses
 - seeking a cooler or warmer environment
 - adjusting clothing
 - changing posture
- Autonomic Responses
 - Sweating
 - Shivering
 - Vasoconstriction/vasodilation
- Endocrine Responses:
 - hormones that increase metabolic heat production

Temperature Set point

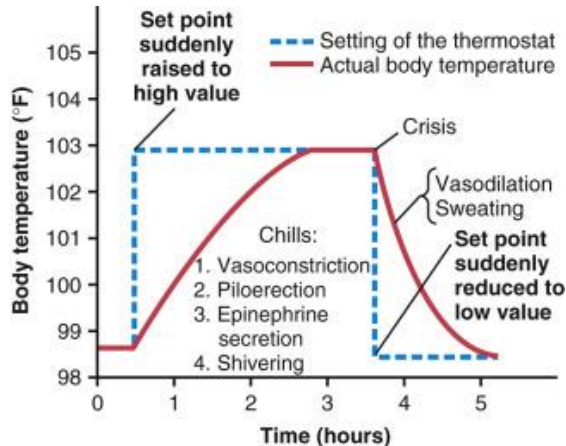
- Temperature set point is a homeostatic hypothalamic mechanism that activates heat production and dissipation to keep the core temperature constant
- Hypothalamic set point is often likened to a central room thermostat
- The set point can be affected by non-thermal signals/factors

- Exercise
- Ovulation
- Pyrogens

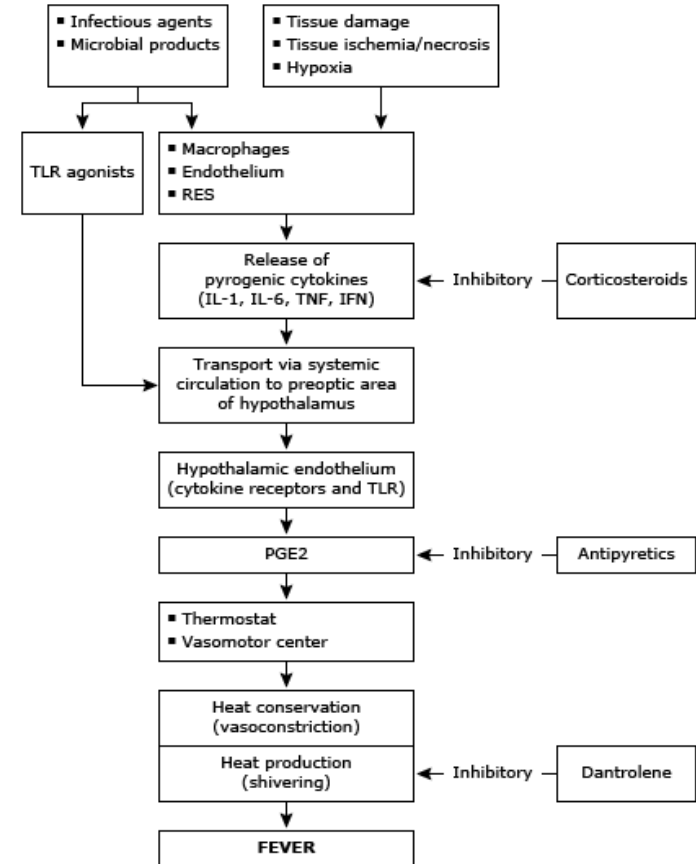


Mechanism of Fever

- Pyrogens – substances that increase the set point
- Symptomology
 - **Chills** – onset of fever - the patient is responding to the elevated setpoint → cold skin (because of vasoconstriction) and shivering
 - **Crisis** – resolution of fever - the factor that is causing the high set point is removed → set point drops to a normal level → cutaneous vasodilation and sweating



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Summary

- The core temperature remains stable
- Skin temperature fluctuates with the environment
- Heat dissipates by thermal radiation, evaporation of sweat, and conduction of heat to the ground and the air
- Cutaneous vasodilation increases heat transfer from the core to the body surface and helps heat dissipation
- The major sources of heat in the body are metabolism and normal physical activity
- Additional heat is produced from the food digestion and thermogenesis
- Thermoregulatory center in the hypothalamus responds to changes in skin and core temperature to activate or suppress neuronal and endocrine regulation of thermogenesis and heat dissipation
- Fever is defined as an increase in hypothalamic “set point” which results in the activation in heat production and reduction of heat dissipation

Lecture Feedback Form:

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>